

STAT 4005 - Time Series

Chapter 7 - Nonstationarity

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Nonstationarity

- Recall the definition of weak stationarity:

$$\left\{ \begin{array}{lcl} \mathbb{E}(Y_t) & = & \mu, \\ \text{Var}(Y_t) & = & \gamma(0), \\ \text{Cov}(Y_t, Y_{t+k}) & = & \gamma(k), \quad k = 1, 2, 3, \dots \end{array} \right. ,$$

independent of time t .

- We have seen in Chapter 3 that $\text{ARIMA}(p, d, q)$ models with $d > 0$ are non-stationary.
- In general, a non-stationary time series may exhibit a systematic change in mean, covariance structure, or both.

Nonstationarity in mean: Random Walk with drift and deterministic time trend

- Two types of nonstationarity in mean:
 - Trend stationary series (A trend with stationary noise)

$$(TS) : Y_t = \beta_0 + \beta_1 t + \nu_t$$

- Differenced stationary series (stationary after differencing)

$$(DS) : Y_t = \beta_1 + Y_{t-1} + \nu_t$$

Nonstationarity in mean: Random Walk with drift and deterministic time trend

- TS and DS require different ways to handle



$$(TS) : Y_t = \beta_0 + \beta_1 t + \nu_t$$

detrending to achieve stationarity



$$(DS) : Y_t = \beta_1 + Y_{t-1} + \nu_t$$

differencing to achieve stationarity

- How to distinguish between the two models by just looking at the data?

Nonstationarity in mean: Random Walk with drift and deterministic time trend

- To distinguish between two models, we build a model that encompasses both situations.

$$Y_t = \beta_0 + \beta_1 t + \nu_t \quad (1)$$

$$\nu_t = \alpha \nu_{t-1} + Z_t \quad (2)$$

- Another representation of the encompassing model
 - From (1), we get

$$\nu_t = Y_t - \beta_0 - \beta_1 t \quad (3)$$

- Substitute (3) into (2) gets

$$Y_t - \beta_0 - \beta_1 t = \alpha(Y_{t-1} - \beta_0 - \beta_1(t-1)) + Z_t$$

$$Y_t = \beta_0(1 - \alpha) + \beta_1\alpha + \beta_1(1 - \alpha)t + \alpha Y_{t-1} + Z_t$$

Nonstationarity in mean: Random Walk with drift and deterministic time trend

- If $\alpha = 1$, the model reduces to

$$(DS) : Y_t = \beta_1 + Y_{t-1} + Z_t.$$

- If $\alpha \neq 1$, the model reduces to

$$(TS) : Y_t = \gamma_0 + \gamma_1 t + \alpha Y_{t-1} + Z_t,$$

where $\gamma_0 = \beta_0(1 - \alpha) + \beta_1\alpha$, $\gamma_1 = \beta_1(1 - \alpha)$.

- This motivates distinguishing between the two models by testing

$$H_0 : \alpha = 1, \quad \text{i.e.,} \quad Y_t = \beta_1 + Y_{t-1} + Z_t,$$

$$H_1 : \alpha < 1, \quad \text{i.e.,} \quad Y_t = \gamma_0 + \gamma_1 t + \alpha Y_{t-1} + Z_t.$$

Nonstationary time series in Finance

- Let P_t = Price of Stock at time t
- Recall that it is common to assume

$$\log P_t - \log P_{t-1} \approx \frac{P_t - P_{t-1}}{P_{t-1}} = \text{white noise} . \quad (4)$$

⇒ If $Y_t = \log P_t$, then $Y_t \sim \text{ARIMA}(0,1,0)$ is non-stationary

- To validate the model in (4) we assume $Y_t - \alpha Y_{t-1} = Z_t$, $Z_t \sim WN(0, \sigma^2)$ and consider the **unit root test**

$$H_0 : \alpha = 1,$$

$$H_1 : \alpha < 1.$$

Unit root test

In general, there are three versions of the unit root test:

- For $Y_t = \alpha Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$, test

$$H_0 : \alpha = 1, \quad \text{v.s.} \quad H_1 : \alpha < 1.$$

- For $Y_t = \beta_0 + \alpha Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$, test

$$H_0 : \alpha = 1, \quad \text{v.s.} \quad H_1 : \alpha < 1.$$

- For $Y_t = \beta_0 + \beta_1 t + \alpha Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$, test

$$H_0 : \alpha = 1, \quad \text{v.s.} \quad H_1 : \alpha < 1.$$

Each version of the test has its own critical value which depends on the size of the sample. For simplicity, we only discuss the first case.

Unit root test

- For $Y_t = \alpha Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$, test

$$H_0 : \alpha = 1,$$

$$H_1 : \alpha < 1.$$

- Idea: Check if the estimated AR coefficient $\hat{\alpha} = \frac{\sum_{t=2}^n Y_t Y_{t-1}}{\sum_{t=2}^n Y_{t-1}^2}$ is close to 1.
- Problem: How close is close? We need to find the distribution of $\hat{\alpha}$.

Unit root test

Basic probabilistic tools to study the distribution of $\hat{\alpha}$:

- $Y_k = \sum_{i=1}^k Z_i$ is a random walk if $Z_i \sim WN(0, 1)$.
- **Functional Central Limit Theorem (FCLT):**
 - A random walk converges to Brownian motion when step size $\downarrow 0$ and number of steps $\uparrow \infty$

$$Y(t) = \frac{Y_{[nt]}}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{[nt]} Z_i \xrightarrow{d} W(t).$$

where $t \in [0, \infty)$ and $W(t)$ is a standard Brownian motion

Remarks:

- Note that $Y(t)$ and $W(t)$ are functions of $t \in [0, \infty)$.
- Ordinary Central Limit Theorem is a special case when $t = 1$:

$$Y(1) = \frac{Y_n}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \xrightarrow{d} W(1) \stackrel{d}{=} N(0, 1).$$

Definition 1

A standard Brownian motion $W(t)$ is a continuous-time stochastic process satisfying the following properties:

- 1 $W(0) = 0$.
- 2 $W(\cdot)$ has independent increments. For any $t_3 \geq t_2 \geq t_1 \geq 0$, $W(t_3) - W(t_2)$ is independent of $W(t_1)$.
- 3 $W(\cdot)$ has Gaussian increments. For any $t_2 \geq t_1 \geq 0$, $W(t_2) - W(t_1)$ is normally distributed with mean 0 and variance $t_2 - t_1$. i.e., $W(t_2) - W(t_1) \sim N(0, t_2 - t_1)$.
- 4 $W(\cdot)$ has continuous sample paths. With probability 1, $W(t)$ is continuous in t .

Unit root test

- FCLT: $Y_k = \sum_{i=1}^k Z_i, \frac{Y_{[nt]}}{\sqrt{n}} \xrightarrow{d} W(t).$

- Unit root statistic: $\hat{\alpha} = \frac{\sum_{t=2}^n Y_t Y_{t-1}}{\sum_{t=2}^n Y_{t-1}^2}$

- Denominator:

$$\sum_{t=2}^n Y_{t-1}^2 = n^2 \sum_{t=2}^n \left(\frac{Y_{t-1}}{\sqrt{n}} \right)^2 \frac{1}{n} \approx n^2 \int_0^1 W^2(t) dt$$

- Numerator:

$$\begin{aligned} \sum_{t=2}^n Y_t Y_{t-1} &= \sum_{t=2}^n Y_{t-1}^2 + n \sum_{t=2}^n \frac{Y_{t-1}}{\sqrt{n}} \frac{Y_t - Y_{t-1}}{\sqrt{n}} \\ &\approx n^2 \int_0^1 W^2(t) dt + n \int_0^1 W(t) dW(t) \end{aligned}$$

- Putting together:

$$n(\hat{\alpha} - 1) \xrightarrow{d} \frac{\int_0^1 W(t) dW(t)}{\int_0^1 W^2(t) dt}$$

Implementation

Unit root test is sometimes called the **Dickey-Fuller Test**

- Setting: for $Y_t = \alpha Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$, test

$$H_0 : \alpha = 1,$$

$$H_1 : \alpha < 1.$$

- Test statistics: $\hat{\alpha} = \frac{\sum_{t=2}^n Y_t Y_{t-1}}{\sum_{t=2}^n Y_{t-1}^2}$.
- Asymptotic distribution of test statistics: $n(\hat{\alpha} - 1) \xrightarrow{d} \frac{\int_0^1 W(t) dW(t)}{\int_0^1 W^2(t) dt}$
- Test procedure: Reject H_0 at 5% significance level if

$$n(\hat{\alpha} - 1) < \text{5-th percentile of } \frac{\int_0^1 W(t) dW(t)}{\int_0^1 W^2(t) dt}.$$

- The percentiles of the distribution $\frac{\int_0^1 W(t) dW(t)}{\int_0^1 W^2(t) dt}$ is tabulated in many books, or can be computed by simulations.
- In R, the Dickey-Fuller test is implemented in the package `tseries`

Example using R

- `library(tseries)`
- `library(quantmod)`
- `x=getSymbols("0005.HK", from="2011-01-01",
to="2014-12-31", src="yahoo", auto.assign=FALSE)`
- `y=as.numeric(log(x[!is.na(x[,6]),6]))`
- `adf.test(y, k=0)`

Note: The results from the `adf.test()` function is based on the version with drift and deterministic time trend.

Nonstationarity in variance

$$Y_t = \mu_t + Z_t, \quad \text{Var}(Z_t) = h^2(\mu_t)\sigma^2$$

- The mean level μ_t is a deterministic function of time t
- Use transformation to "stabilize" the variance
 - Taylor expansion: $g(Y_t) \approx g(\mu_t) + (Y_t - \mu_t)g'(\mu_t)$
 - So, $\text{Var}\{g(Y_t)\} \approx \{g'(\mu_t)\}^2 h^2(\mu_t)\sigma^2$
 - Choosing $g'(\mu_t) = \frac{1}{h(\mu_t)}$, $g(Y_t)$ has constant variance

Nonstationarity in variance

- $Y_t = \mu_t + Z_t, \quad \text{Var}(Z_t) = h^2(\mu_t)\sigma^2$
- $\text{Var}\{g(Y_t)\} = \{g'(\mu_t)\}^2 h^2(\mu_t)\sigma^2$
- Examples
 - $h(\mu_t) = \mu_t$. Then we choose g s.t. $g'(\mu_t) = \frac{1}{\mu_t} \Rightarrow g(\mu_t) = \log(\mu_t)$
 - $h(\mu_t) = \mu_t^{1/2}$. Then we choose g s.t. $g'(\mu_t) = \mu_t^{-1/2} \Rightarrow g(\mu_t) = 2\mu_t^{1/2}$

Nonstationarity in variance

- $Y_t = \mu_t + Z_t, \quad \text{Var}(Z_t) = h^2(\mu_t)\sigma^2$

- Box-Cox transformation

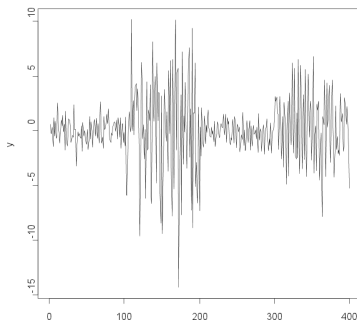
$$g_\lambda(y) = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log y, & \lambda = 0 \end{cases}$$

can be used as a variance-stabilizing transformation. (Trial and error for different λ 's)

- Note that this type of nonstationarity in variance is only one particular case of nonstationarity in variance
- Specifically, $\text{Var}(Z_t)$ is a deterministic function of t .

Heteroskedasticity in time series

- Heteroskedasticity = non-constant variance
- Time Series Context:
 - Variation is small for some periods of time and large for other periods



Heteroskedasticity in time series

- e.g. log assets price series $\log(P_t)$, where P_t = asset price at time t
 - Usually, $X_t = \Delta(\log(P_t)) = \epsilon_t$ is treated as white noises
 - i.e. $\log(P_t) \sim \text{ARIMA}(0,1,0)$
 - Empirically, ϵ_t is found to have different variance over time
- R Codes:
 - `x=getSymbols("0002.HK", from="2010-01-01", to="2016-12-31", src="yahoo", auto.assign=FALSE)`
 - `y=as.numeric(diff(log(x[!is.na(x[,6]),6])))[-1]`
 - `par(mfrow=c(3,1))`
 - `ts.plot(y); acf(y); acf(y^2)`

Stylized facts for log asset series X_t

- In reality
 - ① X_t is heavy-tailed (probability of taking an extreme value is higher than that implied by the normal distribution)
 - ② $\{X_t\}$ is not serially correlated, but $\{X_t^2\}$ is serially correlated
 - ③ Volatility cluster: large (small) changes in $\{X_t\}$ tend to be followed by large (small) changes
- These are the **stylized facts** of financial data

AutoRegressive Conditional Heteroskedastic Model (ARCH)

- A very popular model for volatility
- Model: ARCH(q)

$$\begin{aligned}X_t &= \sigma_t \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, 1) \\ \sigma_t^2 &= \alpha_0 + \sum_{i=1}^q \alpha_i X_{t-i}^2\end{aligned}$$

- Meaning of conditional heteroskedastic model
 - Let $\mathcal{F}_{t-1} = \sigma(X_{t-1}, X_{t-2}, \dots)$ be the σ -field (information) generated by all past information up to time $t-1$

$$\text{Var}(X_t | \mathcal{F}_{t-1}) = \text{E}(X_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2 \text{E}(\epsilon_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i X_{t-i}^2$$

- Conditional variance depends on previous values of X_t^2 's
- Restrictions for the parameter set $(\alpha_0, \alpha_1, \dots, \alpha_q)$
 - $\alpha_i \geq 0, \quad i = 0, 1, \dots, q$: Variance must be positive
 - $\alpha_1 + \dots + \alpha_q < 1$: Make sure σ_t^2 is stationary, otherwise σ_t^2 explodes

Example: ARCH(1) model

- ARCH(1) model:

$$X_t = \sigma_t \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, 1), \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2.$$

- If $\alpha_1 < 1$, then

$$\begin{aligned} X_t^2 &= \epsilon_t^2 (\alpha_0 + \alpha_1 \epsilon_{t-1}^2 \sigma_{t-1}^2) \\ &= \alpha_0 \epsilon_t^2 + \alpha_0 \alpha_1 \epsilon_t^2 \epsilon_{t-1}^2 + \alpha_1^2 \epsilon_t^2 \epsilon_{t-1}^2 \epsilon_{t-2}^2 \sigma_{t-2}^2 = \dots \\ &= \alpha_0 \sum_{j=0}^{\infty} \alpha_1^j \epsilon_t^2 \dots \epsilon_{t-j}^2. \end{aligned}$$

- X_t is **causal** but **nonlinear**: not in the form $X_t = \sum_{i=0}^{\infty} \psi_i \epsilon_{t-i}$
- Two methods to find $E(X_t^2)$
 - $E(X_t^2) = \alpha_0 \sum_{j=0}^{\infty} \alpha_1^j = \frac{\alpha_0}{1-\alpha_1}$ ($E(\epsilon_t^2) = 1, \forall t$)
 - Take expectation on both sides on $X_t^2 = \epsilon_t^2 (\alpha_0 + \alpha_1 X_{t-1}^2)$, then

$$E(X_t^2) = \alpha_0 E(\epsilon_t^2) + \alpha_1 E(X_{t-1}^2)$$

$$\Rightarrow E(X_t^2) = \frac{\alpha_0}{1-\alpha_1} \text{ as } E(X_t^2) = E(X_k^2) \text{ for all pairs of } t, k$$

Example: ARCH(1) model

Properties of ARCH(1)

- X_t is causal and non-linear
-

$$E(X_t) = E(\sigma_t \epsilon_t) = E(\sigma_t)E(\epsilon_t) = 0,$$

$$\text{Var}(X_t) = E(X_t^2) = \frac{\alpha_0}{1 - \alpha_1},$$

$$\begin{aligned}\text{Cov}(X_{t+h}, X_t) &= E(X_{t+h} X_t) \quad (E(X_t) = 0) \\ &= E(\sigma_{t+h} \sigma_t \epsilon_{t+h} \epsilon_t) \\ &= E(\sigma_{t+h} \sigma_t \epsilon_t) E(\epsilon_{t+h}) \\ &= 0.\end{aligned}$$

- Conditional variance $\neq$ unconditional variance

$$\text{Var}(X_t | \mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 \neq \text{Var}(X_t) = \frac{\alpha_0}{1 - \alpha_1}$$

Example: ARCH(1) model

- Connection between ARCH(1) and AR(1)

$$\text{ARCH}(1) : X_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$$

- Rewrite

$$\begin{aligned} X_t^2 &= \sigma_t^2 + X_t^2 - \sigma_t^2 \\ &= \alpha_0 + \alpha_1 X_{t-1}^2 + \sigma_t^2 (\epsilon_t^2 - 1) \\ &= \alpha_0 + \alpha_1 X_{t-1}^2 + \nu_t \end{aligned}$$

- If $X_t \sim \text{ARCH}(1)$, then $X_t^2 \sim \text{AR}(1)$ with **noise** $\nu_t = \sigma_t^2 (\epsilon_t^2 - 1)$

Example: ARCH(1) model

Properties of $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$

$$\begin{aligned}E(\nu_t) &= E(\sigma_t^2(\epsilon_t^2 - 1)) \\&= E(\sigma_t^2)E(\epsilon_t^2 - 1) \\&= 0\end{aligned}$$

$$\begin{aligned}\text{Var}(\nu_t) &= E(\nu_t^2) \\&= E(\sigma_t^4(\epsilon_t^2 - 1)^2) \\&= E(\sigma_t^4)E(1 - 2\epsilon_t^2 + \epsilon_t^4) \\&= 2E(\sigma_t^4) \quad (\epsilon_t \stackrel{\text{iid}}{\sim} N(0, 1) \Rightarrow E(\epsilon_t^4) = 3)\end{aligned}$$

$$\begin{aligned}\text{Cov}(\nu_t, \nu_{t+h}) &= E(\sigma_t^2 \sigma_{t+h}^2 (\epsilon_t^2 - 1)(\epsilon_{t+h}^2 - 1)) \\&= E(\sigma_t^2 \sigma_{t+h}^2 (\epsilon_t^2 - 1))E(\epsilon_{t+h}^2 - 1) \\&= 0\end{aligned}$$

Thus, $\{\nu_t\}$ is indeed a white noise sequence.

Example: ARCH(1) model

ARCH(1) : $X_t = \sigma_t \epsilon_t$, $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$

AR(1) representation: $X_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \nu_t$, where $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$.

- The AR(1) representation can be regarded as the description of the evolution of the variance of the process $\{X_t\}$
 - $X_t^2 = \sigma_t^2 \epsilon_t^2 \approx \sigma_t^2$, $(E(\epsilon_t^2) = 1)$
- The correlation structure of X_t^2 follows the same calculation of AR(1) model

$$\text{Var}(X_t^2) = \frac{\text{Var}(\nu_t)}{1 - \alpha_1^2},$$

$$\text{Cov}(X_t^2, X_{t+h}^2) = \frac{\alpha_1^{|h|} \text{Var}(\nu_t)}{1 - \alpha_1^2},$$

$$\text{Corr}(X_t^2, X_{t+h}^2) = \alpha_1^{|h|}$$

Example: ARCH(1) model

- To completely determine the covariance structure of $\{X_t^2\}$ it remains to compute $\text{Var}(\nu_t) = 2E(\sigma_t^4)$
- Square both sides of $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2$ and take expectation

$$\begin{aligned} E(\sigma_t^4) &= \alpha_0^2 + 2\alpha_0\alpha_1 E(X_{t-1}^2) + \alpha_1^2 E(X_{t-1}^4) \\ &= \alpha_0^2 + 2\alpha_0\alpha_1 \frac{\alpha_0}{1 - \alpha_1} + 3\alpha_1^2 E(\sigma_t^4) \\ \Rightarrow E(\sigma_t^4) &= \frac{\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)} \end{aligned}$$

- Note that the 4th moment $E(\sigma_t^4)$ exists only if $\alpha_1^2 < \frac{1}{3}$

The Usefulness of ARCH(1) Model

ARCH models successfully captures the three stylized facts.

Stylized fact 1: X_t is heavy-tailed

- Usually, the heavier the tail, the larger the 4th moment

$$E(X_t^4) = E(\sigma_t^4)E(\epsilon_t^4) = \frac{3\alpha_0(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)}$$

- If α_1 is close to $1/\sqrt{3}$, then $E(X_t^4)$ can be very large
- The α_1 is found to be around 0.9 in many financial time series, i.e., 4-th moment does not exist.

The Usefulness of ARCH(1) Model

Stylized fact 2: Correlated structure

- $\{X_t\}$ is not serially correlated
 $\{X_t^2\}$ is non-negatively serially correlated
- This means
$$\text{Cov}(X_t, X_{t+h}) = 0, \text{ for } h \neq 0$$
$$\text{Corr}(X_t^2, X_{t+h}^2) = \alpha_1^{|h|} > 0 \text{ as } \alpha_1 > 0 \text{ by assumption}$$

The Usefulness of ARCH(1) Model

Stylized fact 3: Volatility cluster

- Large (small) changes in $\{X_t\}$ tend to be followed by large (small) changes
- Well explained by the equation

$$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2, \quad \text{as } \alpha_1 > 0$$

Generalized ARCH (GARCH) model

- GARCH(p, q) model

$$\begin{aligned}X_t &= \sigma_t \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, 1) \\ \sigma_t^2 &= \alpha_0 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + \sum_{j=1}^q \alpha_j X_{t-j}^2\end{aligned}$$

- Parameter set: $(\alpha_0, \alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p)$
 - higher flexibility for modelling
- Constraints

$$\begin{aligned}\alpha_j &\geq 0, \quad j = 0, \dots, q, \\ \beta_i &\geq 0, \quad i = 1, \dots, p, \\ \alpha_1 + \dots + \alpha_q + \beta_1 + \dots + \beta_p &< 1\end{aligned}$$

GARCH and ARMA

- GARCH(1,1) is the most commonly used GARCH model in practice

$$X_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

- Connection to ARMA(1,1) model:

$$\begin{aligned} X_t^2 &= \sigma_t^2 + X_t^2 - \sigma_t^2 \\ &= \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + X_t^2 - \sigma_t^2 \\ &= \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 X_{t-1}^2 - \beta_1 (X_{t-1}^2 - \sigma_{t-1}^2) + X_t^2 - \sigma_t^2 \\ &= \alpha_0 + (\alpha_1 + \beta_1) X_{t-1}^2 + \nu_t - \beta_1 \nu_{t-1}, \\ &\sim \text{ARMA}(1, 1) \text{ with white noise } \{\nu_t\}, \end{aligned}$$

$$\text{where } \nu_t = X_t^2 - \sigma_t^2 = \sigma_t^2(\epsilon_t^2 - 1)$$

Theorem 2

If X_t is a GARCH(p, q) process, then X_t^2 is an ARMA(m, p) process with noise $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$, where $m = \max(p, q)$.

Exercise

Let X_t follows a GARCH(2,3) model,

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^2 \beta_i \sigma_{t-i}^2 + \sum_{j=1}^3 \alpha_j X_{t-j}^2.$$

Show that X_t^2 can be written as an ARMA(3,2) model in terms of the process $\nu_t = \sigma_t^2(\epsilon_t^2 - 1)$. Identify the parameters of the ARMA process in terms of the parameters of the given GARCH(2,3) model.

Solution

Let X_t be a GARCH(2,3) model satisfying

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^2 \beta_i \sigma_{t-i}^2 + \sum_{j=1}^3 \alpha_j X_{t-j}^2.$$

Then,

$$\begin{aligned} X_t^2 &= \sigma_t^2 + (X_t^2 - \sigma_t^2) \\ &= \alpha_0 + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \alpha_1 X_{t-1}^2 + \alpha_2 X_{t-2}^2 + \alpha_3 X_{t-3}^2 \\ &\quad + X_t^2 - \sigma_t^2 \\ &= \alpha_0 + (\alpha_1 + \beta_1) X_{t-1}^2 + (\alpha_2 + \beta_2) X_{t-2}^2 + \alpha_3 X_{t-3}^2 \\ &\quad - \beta_1 (X_{t-1}^2 - \sigma_{t-1}^2) - \beta_2 (X_{t-2}^2 - \sigma_{t-2}^2) + X_t^2 - \sigma_t^2 \\ &= \alpha_0 + (\alpha_1 + \beta_1) X_{t-1}^2 + (\alpha_2 + \beta_2) X_{t-2}^2 + \alpha_3 X_{t-3}^2 \\ &\quad - \beta_1 \nu_{t-1} - \beta_2 \nu_{t-2} + \nu_t, \end{aligned}$$

where $\nu_t = X_t^2 - \sigma_t^2 = \sigma_t^2(\epsilon_t^2 - 1)$.

Now, we can identify the equation of an ARMA(3,2) model with noise ν_t .

- For GARCH(1,1) model with $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ to be stationary, one necessary condition is

$$\alpha_1 + \beta_1 < 1.$$

- In practice, $\alpha_1 + \beta_1 \approx 1$ is observed in many real datasets
- When $\alpha_1 + \beta_1 = 1$, the process is nonstationary and is known as Integrated GARCH (I-GARCH) process.

I-GARCH model

- $\alpha_1 + \beta_1 = 1$:

$$\begin{aligned}E(\sigma_{t+1}^2 \mid \mathcal{F}_{t-1}) &= E(\alpha_0 + \alpha_1 X_t^2 + \beta_1 \sigma_t^2 \mid \mathcal{F}_{t-1}) \\&= \alpha_0 + (\alpha_1 + \beta_1) \sigma_t^2 \\&= \alpha_0 + \sigma_t^2, \\E(\sigma_{t+2}^2 \mid \mathcal{F}_{t-1}) &= E(E(\sigma_{t+2}^2 \mid \mathcal{F}_t) \mid \mathcal{F}_{t-1}) \\&= E(\alpha_0 + \sigma_{t+1}^2 \mid \mathcal{F}_{t-1}) \\&= 2\alpha_0 + \sigma_t^2.\end{aligned}$$

- Similarly, $E(\sigma_{t+j}^2 \mid \mathcal{F}_{t-1}) = j\alpha_0 + \sigma_t^2$.
 - Thus, I-GARCH is non-stationary
 - The volatility is persistent: For $j > 0$, $E(\sigma_{t+j}^2 \mid \mathcal{F}_{t-1})$ depends on σ_t^2
 - For GARCH(1,1) with $\alpha_1 + \beta_1 < 1$,
 $E(\sigma_{t+j}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 \sum_{i=0}^{j-1} (\alpha_1 + \beta_1)^i + (\alpha_1 + \beta_1)^j \sigma_t^2$ depends on σ_t^2 with diminishing effect.

Exercise

Show that for an I-GARCH(1,1) model, for $j > 0$,

$$\mathbb{E}(\sigma_{t+j}^2 | \mathcal{F}_{t-1}) = j\alpha_0 + \sigma_t^2.$$

Solution:

$$\begin{aligned}\mathbb{E}(\sigma_{t+j}^2 | \mathcal{F}_{t-1}) &= \mathbb{E}[\mathbb{E}(\sigma_{t+j}^2 | \mathcal{F}_{t+j-2}) | \mathcal{F}_{t-1}] \\ &= \alpha_0 + \mathbb{E}(\sigma_{t+j-1}^2 | \mathcal{F}_{t-1}) \\ &= 2\alpha_0 + \mathbb{E}(\sigma_{t+j-2}^2 | \mathcal{F}_{t-1}) \\ &= \dots \\ &= j\alpha_0 + \mathbb{E}(\sigma_t^2 | \mathcal{F}_{t-1}) \\ &= j\alpha_0 + \sigma_t^2,\end{aligned}$$

since σ_t^2 is $\mathcal{F}_{t-1}$ measurable.

Further generalization of GARCH model

- t-GARCH
 - Use t -distribution instead of Normal for noise sequence ϵ_t
 - Capture more heavy-tail features
- e-GARCH
 - Use the exponential function to model the asymmetry
- Others: n-GARCH, GARCH.M, ...

Test for heteroskedasticity

- Testing for heterogeneity: **Lagrange multiplier test** (LM)

Theorem 3

If $X_t \sim ARCH(q)$, then for the fitted regression equation

$$\hat{X}_t^2 = \hat{\alpha}_0 + \sum_{i=1}^q \hat{\alpha}_i X_{t-i}^2,$$

the coefficient of determination, R^2 , satisfies $nR^2 \rightarrow \chi_q^2$ under H_0 : Homoscedasticity when the sample size, n , is large.

- If $p\text{-value} < 0.05$, then H_0 is rejected, and we conclude that X_t is heteroskedastic.
- $R^2 = \frac{\sum_{t=1}^n (\hat{X}_t^2 - \bar{X}^2)^2}{\sum_{t=1}^n (X_t^2 - \bar{X}^2)^2}$ = proportion of variation explained by the regression model

Test for heteroskedasticity

Lagrange multiplier (LM) test

- Fit $\hat{X}_t^2 = \hat{\alpha}_0 + \sum_{i=1}^q \hat{\alpha}_i X_{t-i}^2$.
- Reject H_0 if

$$\Pr(\chi_q^2 > nR^2) < 0.05.$$

Example:

- `x=getSymbols("0002.HK", from="2011-01-01", to="2014-12-31", src="yahoo", auto.assign=FALSE)`
- `y=as.numeric(diff(log(x[!is.na(x[,6]),6])))[-1]`
- `y2=y^2`
- `n=length(y)`
- `fit=lm(y2[2:n]~y2[1:(n-1)])`
- `R2=summary(fit)$r.squared`
- `p.value=1-pchisq(n*R2,1)`

Estimation of ARCH model

- ARCH(q): $X_t = \sigma_t \epsilon_t$, $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \dots + \alpha_q X_{t-q}^2$
- Conditional on $\mathcal{F}_{t-1}$, σ_t^2 is known. Hence, we have $X_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$
- Conditional probability density:

$$f(x_t | \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_t^2}} e^{-\frac{1}{2\sigma_t^2} x_t^2}.$$

Estimation of ARCH model

- Likelihood of ARCH(q):

$$f(x_n, \dots, x_1 \mid \mathcal{F}_0) = f(x_n \mid x_{n-1}, \dots) \cdots f(x_2 \mid x_1, \dots) f(x_1 \mid x_0, \dots)$$

$$\begin{aligned} \log\{f(x_n, \dots, x_1 \mid \mathcal{F}_0)\} &= \sum_{t=1}^n \log f(x_t \mid \mathcal{F}_{t-1}) \\ &= \sum_{t=1}^n \log \frac{1}{\sqrt{2\pi\sigma_t^2}} e^{-\frac{1}{2\sigma_t^2} x_t^2} \\ &= -\frac{n}{2} \log(2\pi) - \sum_{t=1}^n \frac{1}{2} \log \sigma_t^2 - \frac{1}{2} \sum_{t=1}^n \frac{x_t^2}{\sigma_t^2} \end{aligned}$$

- Substitute $\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i x_{t-i}^2$ (with $x_s = 0$ for $s \leq 0$), the MLE can be obtained by maximizing the log-likelihood numerically.

Estimation of GARCH model

Likelihood of GARCH(p, q):

- In fact, the same argument holds:

$$\log\{f(x_n, \dots, x_1 \mid \mathcal{F}_0)\} = -\frac{n}{2} \log(2\pi) - \sum_{t=1}^n \frac{1}{2} \log \sigma_t^2 - \frac{1}{2} \sum_{t=1}^n \frac{x_t^2}{\sigma_t^2}$$

- The only difference is that σ_t^2 is not observed explicitly
- Use recursion to obtain the approximate version of σ_t^2 , denoted as $\tilde{\sigma}_t^2$

$$\tilde{\sigma}_1^2 = \alpha_0$$

$$\tilde{\sigma}_2^2 = \alpha_0 + \alpha_1 x_1^2 + \beta_1 \tilde{\sigma}_1^2$$

$$\tilde{\sigma}_3^2 = \alpha_0 + \alpha_1 x_2^2 + \alpha_2 x_1^2 + \beta_1 \tilde{\sigma}_2^2 + \beta_2 \tilde{\sigma}_1^2$$

$$\vdots$$

$$\tilde{\sigma}_n^2 = \alpha_0 + \sum_{i=1}^q \alpha_i x_{n-i}^2 + \sum_{j=1}^p \beta_j \tilde{\sigma}_{n-j}^2$$

- R: `garch(x, order=c(p,q))`

Model selection for GARCH model

- To select the order q for ARCH model
 - Use the AR representation and use PACF plot to identify q
- More difficult for GARCH
 - $\text{GARCH}(p, q)$ corresponds to $\text{ARMA}(m, p)$, $m = \max(p, q)$.
 - Information on q is lost
 - ⇒ Difficult to detect the order of ARMA using (P)ACF plot
- Use AIC/BIC

Model selection for GARCH model

Example:

- `set.seed(4005); n=300; burnin=100; x=rep(0,n+burnin); e=rnorm(n+burnin)`
- `for(i in 3:(n+burnin)){ # Generate ARCH(2) process
 $x[i] = e[i] \cdot \sqrt{1 + 0.4 \cdot x[i-1]^2 + 0.5 \cdot x[i-2]^2}$
}`
- `x=x[-c(1:burnin)]`
- `pacf(x^2)`
- `ans=rep(0,8)`
- `ans[1]=AIC(garch(x,c(1,0)))`
- `ans[2]=AIC(garch(x,c(0,1)))`
- `ans[3]=AIC(garch(x,c(2,0)))`
- `ans[4]=AIC(garch(x,c(0,2)))`
- `ans[5]=AIC(garch(x,c(1,1)))`
- `ans[6]=AIC(garch(x,c(1,2)))`
- `ans[7]=AIC(garch(x,c(2,1)))`
- `ans[8]=AIC(garch(x,c(2,2)))`

Goodness of fit test for GARCH model

- Goodness of fit: **Ljung-Box test** for residuals:

$$Q(h) = n(n+2) \sum_{j=1}^h \frac{r^2(j)}{n-j} \rightarrow \chi_{h-p-q-1}^2$$

- $r(j)$ is the ACF of the residuals of the GARCH(p, q) model
- R Codes
 - `x=getSymbols("0002.HK", from="2011-01-01", to="2014-12-31", src="yahoo", auto.assign=FALSE)`
 - `y=as.numeric(diff(log(x[!is.na(x[,6]),6])))[-1]`
 - `fit=garch(y,order=c(1,1))`
 - `summary(fit)`
 - `z=fit$residuals[!is.na(fit$residuals)]`
 - `par(mfrow=c(2,2))`
 - `acf(y);acf(y^2);acf(z);acf(z^2)`

Foreign exchange rates example

- Weekly exchange rates of US dollar and British pound between 1980-1988 (exchange.dat)
- Exploratory data analysis:

```
ex.s=scan("exchange.dat")
dex=diff(ex.s)
par(mfrow=c(3,2))
ts.plot(ex.s)
acf(ex.s)
ts.plot(dex)
acf(dex)
ts.plot(dex*dex)
acf(dex*dex)
```

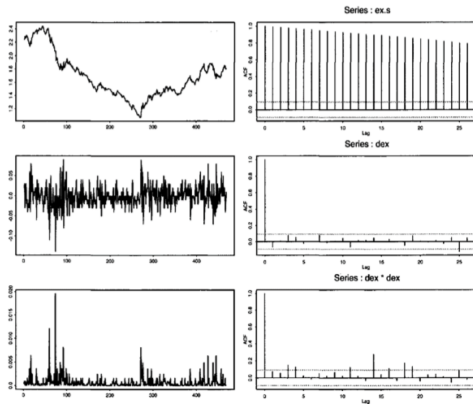


Fig. 9.2 Time series plots and ACFs for X_t and X_t^2 .

Foreign exchange rates example

- Test for heteroskedasticity: Lagrange Multiplier test

```
ex.s=scan("exchange.dat")
dex=diff(ex.s)
x1=c(0,dex[1:468])
x2=c(0,0,dex[1:467])
x3=c(0,0,0,dex[1:466])
x4=c(0,0,0,0,dex[1:465])
z1=x1*x1; z2=x2*x2; z3=x3*x3; z4=x4*x4
lm.1=lm(dex*dex~z1+z2+z3+z4)
summary(lm.1)
t=length(dex)*summary(lm.1)$r.squared
1-pchisq(t,4)
[1] 0.000829048
```

- Conclusion: $p\text{-value} = 0.00083 < 0.05$, the presence of heterogeneity is confirmed

Foreign exchange rates example

```
> summary(lm.1)
```

```
Call:
```

```
lm(formula = dex * dex ~ z1 + z2 + z3 + z4)
```

```
Residuals:
```

	Min	1Q	Median	3Q	Max
	-0.0028480	-0.0006073	-0.0004331	0.0001337	0.0187001

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.113e-04	9.319e-05	5.486	6.76e-08	***
z1	4.693e-02	4.614e-02	1.017	0.30963	
z2	3.328e-02	4.575e-02	0.727	0.46729	
z3	1.379e-01	4.576e-02	3.013	0.00273	**
z4	1.099e-01	4.615e-02	2.381	0.01766	*

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.00151 on 464 degrees of freedom
```

```
Multiple R-squared: 0.04026,    Adjusted R-squared: 0.03199
```

```
F-statistic: 4.866 on 4 and 464 DF,  p-value: 0.0007485
```

Foreign exchange rates example: Model Fitting

```
>library(tseries)
>dex.mod=garch(dex,order=c(1,1))
>summary(dex.mod)
```

Call:

```
garch(x = dex, order = c(1, 1))
```

Model:

```
GARCH(1,1)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.4816	-0.6675	0.0000	0.5410	4.3634

Coefficient(s):

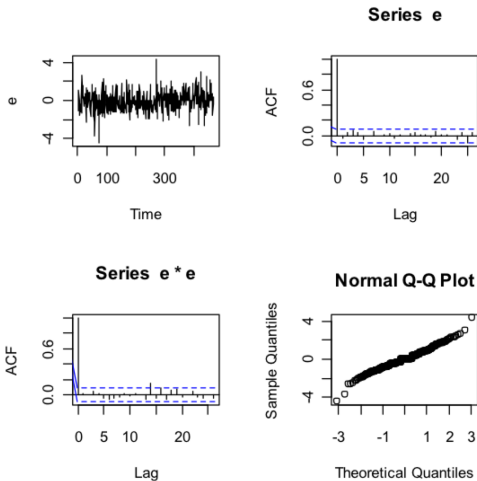
	Estimate	Std. Error	t value	Pr(> t)
a0	3.069e-05	1.356e-05	2.262	0.02369 *
a1	6.638e-02	2.040e-02	3.254	0.00114 **
b1	8.932e-01	3.339e-02	26.748	< 2e-16 ***

- Fitted model is:

$$\sigma_t^2 = 3.1 \times 10^{-5} + 0.066X_{t-1}^2 + 0.89\sigma_{t-1}^2.$$

Foreign exchange rates example

```
ex.s=scan("exchange.dat")
dex=diff(ex.s)
library(tseries)
dex.mod=garch(dex,order=c(1,1))
e=dex.mod$residuals
e=e[-1]      # (e[1]=NA)
par(mfrow=c(2,2))
ts.plot(e)
acf(e)
acf(e*e)
qqnorm(e)
```



Exercise

Perform a GARCH analysis on the U.S. Treasury bill data set

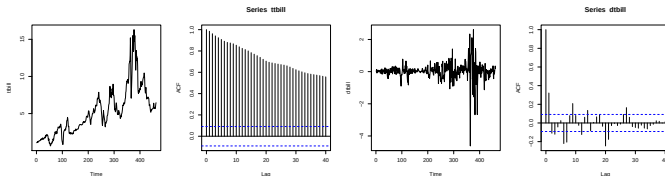
ustbill.dat in

<http://www.sta.cuhk.edu.hk/NHCHAN/TSBook2nd/dataset.html>.

Solution I

- a) From the time series plot and the ACF plot it is clear that the series is not stationary. A differencing might be appropriate in this case. The differenced series looks stationary but the ACF shows several significant lag correlations (6,7,9,12,14,20,21,27).

```
ttbill=read.table("ustbill.dat")
ttbill=ttbill[,-1]; ttbill=as.vector(t(ttbill))
par(mfrow=c(1,4)); ts.plot(ttbill); acf(ttbill,40)
dtbill=diff(ttbill); ts.plot(dtbill); acf(dtbill,40)
```

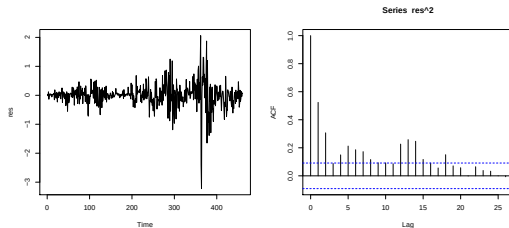


Solution II

For simplicity we consider an AR(27) model. The residual plot of the AR(27) model shows some clusters of large residuals and clusters of small residuals. Moreover, the ACF plot of the squared residuals indicates that serial correlation exists. These can be interpreted as nonconstant variance and would thus fit into the assumptions of GARCH model.

```
fit=arima(dtbill,order=c(27,0,0))
res=fit$residuals
par(mfrow=c(1,2))
ts.plot(res^2);acf(res^2)
```

Solution III



- 9 We fit a $\text{GARCH}(p,q)$ model to the residual series for $p, q \leq 5$ and compare the AIC:

```
ans=matrix(0,6,6)
for (i in 0:5){
  for (j in 0:5){
    if ((i==0)&(j==0)){ ans[i+1,j+1]=0}
    else{ ans[i+1,j+1]=AIC(garch(res,order=c(i,j)))}
  }
}
```

Solution IV

It is found that GARCH(1,1) gives the smallest AIC of 153.32.

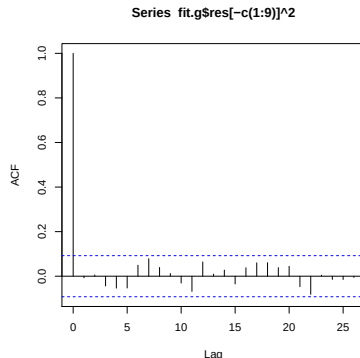
- e) Looking closely into the GARCH(1,1) model. The fitted model is

$$e_t = \sigma_t \eta_t; \quad \sigma_t^2 = 0.0020247 + 0.2580783\sigma_{t-1}^2 + 0.7592271e_{t-1}^2.$$

where e_t is the residual of the AR(27) model and $\eta_t \sim N(0, 1)$. The Ljung-Box test (from `summary(fit.g)`) does not reject the null hypothesis, suggesting that the fitting is appropriate. Moreover, ACF plot of the squared residuals of the GARCH fitting shows no serial correlations, indicating that the non-constant variance effect has been removed.

```
fit.g=garch(res,order=c(1,1))
summary(fit.g)
acf(fit.g$res[-c(1:9)]^2)    # The first 9 NA entries should be
removed.
```

Solution V



- ➊ To conclude, an $\text{ARIMA}(27,1,0)\text{-GARCH}(1,1)$ is proposed to adequately explain the Treasury Bill data.